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SYSTEM DESIGN SPECIFICATION

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1.0 Data Dictionary

1.1 User Database

Description: Stores information about the users for Booking.com

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
UserID	Unique identifier for user	Integer	N/A	Primary Key, Auto-increment	N/A	Booking (One to Many)
FirstName	First name of user	String	30	Not Null	N/A	N/A
LastName	LastName of User	String	30	Not Null	N/A	N/A
Email	Email address of user	String	80	Not Null, Unique	N/A	N/A
PhoneNumber	Phone number of User	Integer	20	Not Null	N/A	N/A
City	Place of user	String	100	Not Null	N/A	N/A

1.2 Booking Database

Description: Booking details from users in Booking.com

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
BookingID	Unique identifier for each Booking details	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Payment)
BookingType	Selected type of booking (hotels, flights, cars, etc.)	String	30	Not Null	N/A	N/A
BookingStatus	Status of the booking	String	30	Not Null	Pending	N/A
PaymentStatus	Whether the payment has been made	String	N/A	Not Null	Unpaid	N/A
ConfirmedAT	Date and time the booking was made	DateTime	N/A	Not Null	N/A	N/A
Duration	Duration of the booking	DateTime	N/A	Not Null	N/A	N/A
TotalAmount	Total cost of booking	Decimal	N/A	Not Null	0.00	N/A
UserID	ID of the user who made the booking	Integer	N/A	Foreign Key (User)	N/A	Many to one (User)
HotelID	FK to Hotel table	Integer	N/A	Foreign Key (Hotel)	NULL	One to One (Hotel)
FlightID	FK to Flight table	Integer	N/A	Foreign Key (Flight)	NULL	One to One (Flight)

BusID	FK to Bus table	Integer	N/A	Foreign Key (Bus)	NULL	One to One (Bus)
RentalCarID	FK to RentalCar table	Integer	N/A	Foreign Key (RentalCar)	NULL	One to One (RentalCar)
TourGuideID	FK to TourGuide table	Integer	N/A	Foreign Key (TourGuide)	NULL	One to One (TourGuide)

1.3 Hotel Database

Description: Information of the Hotels listed in Booking.com

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
HotelID	Unique identifier for hotel	Integer	N/A	Primary Key, Auto-increment	N/A	One to Many (Booking)
HotelName	Name of hotel	String	80	Not Null	N/A	N/A
Location	City or Area of hotel	String	100	Not Null	N/A	N/A
PricePerNight	Rate per night	Decimal	N/A	Not Null	0.00	N/A
MaxGuest	Max number of guest	Integer	N/A	Not Null	N/A	N/A
Rating	Guest rating	Decimal	N/A	Nullable	Null	N/A
Description	Hotel facilities description	String	N/A	Nullable	Null	N/A

1.4 Bus Database

Description: Information of the Bus listed in Booking.com

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
BusID	Unique identifier for bus	Integer	N/A	Primary Key, Auto-increment	N/A	One to Many (Booking)
Name	Name of bus company	String	100	Not Null	N/A	N/A
FromLocation	Departure City	String	100	Not Null	N/A	N/A
ToLocation	Destination city	String	100	Not Null	N/A	N/A
DepartureTime	Bus departure time	Datetime	N/A	Not Null	N/A	N/A
ArrivalTime	Bus arrival time	Datetime	N/A	Not Null	N/A	N/A
Price	Ticket price	Decimal	N/A	Not Null	0.00	N/A
SeatsAvailability	Available seats	Integer	N/A	Not Null	N/A	N/A

1.5 RentalCar Database

Description: Information of the rental cars listed in Booking.com

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
CarRentalID	Unique identifier for car	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
CompanyName	Car Rental company name	String	100	Not Null	N/A	N/A
CarModel	Model of car	String	100	Not Null	N/A	N/A
Location	Pickup Location	String	100	Not Null	N/A	N/A
PricePerDay	Daily rental cost	Decimal	N/A	Not Null	0.00	N/A
Availability	Availability status	Boolean	N/A	Not Null	True	N/A

1.5 TourGuide Database

Description: Information of the Tour guides listed in Booking.com

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
GuideID	Unique identifier for tour guide	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
Name	Name of guide	String	100	Not Null	N/A	N/A
Language	Languages spoken	String	100	Not Null	N/A	N/A
TourType	Tour Categories	String	100	Not Null	N/A	N/A
Rating	Average user rating	Decimal	N/A	Nullable	Null	N/A

PricePerDay	Cost per day of service	Decimal	N/A	Not Null	0.00	N/A
Description	Tour Guide profile	String	100	Nullable	Null	N/A

1.6 Flight Database

Description: Information of the flight listed in Booking.com

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
FlightID	Unique identifier for flights	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
AirlineName	Name of Airline	String	100	Not Null	N/A	N/A
DepartureCity	City of departure	String	100	Not Null	N/A	N/A
ArrivalCity	Destination city	String	100	Not Null	N/A	N/A
DepartureTime	Flight departure time	Datetime	N/A	Not Null	N/A	N/A
ArrivalTime	Flight arrival time	Datetime	N/A	Not Null	N/A	N/A
Price	Ticket cost	Decimal	N/A	Not Null	0.00	N/A
SeatAvailability	Available seats	Integer	N/A	Not Null	0	N/A

1.7 Payment Database

Description: Payment Infomation for the users

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships
PaymentID	Unique identifier for the payment	Integer	N/A	Primary Key, Auto-increment	N/A	One to One (Booking)
BookingID	Asscociated booking being paid for	Integer	N/A	Foreign Key (Booking)	N/A	One to One (each booking once)
TotalAmount	Total cost paid	Decimal	N/A	Not Null	0.00	N/A
PaymentMethod	Method used (Creditcard, Debitcard, etc.)	String	50	Not Null	N/A	N/A
Status	Payment status	String	20	Not Null	Pending	N/A
PaidDuring	Timestamp of successful payment	Datetime	N/A	Not Null	N/A	N/A
UserID	Bill for the user	int	N/A	Not Null	N/A	One to One (user)

1.8 TripPlannerAI

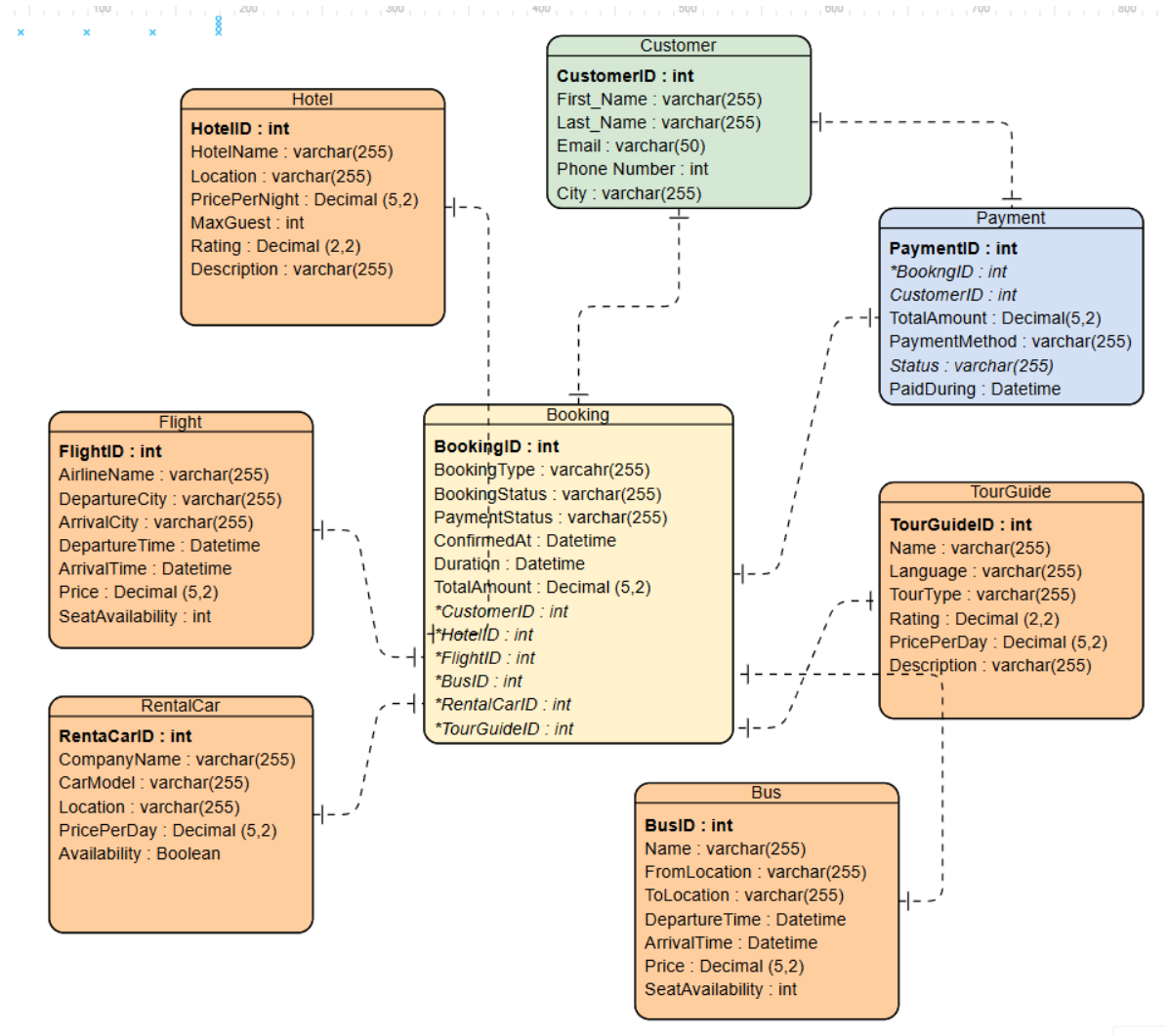
Description: Details for the AI used to plan trips for users

Column Name	Description	Data Type	Length	Constraints	Default Value	Relationships

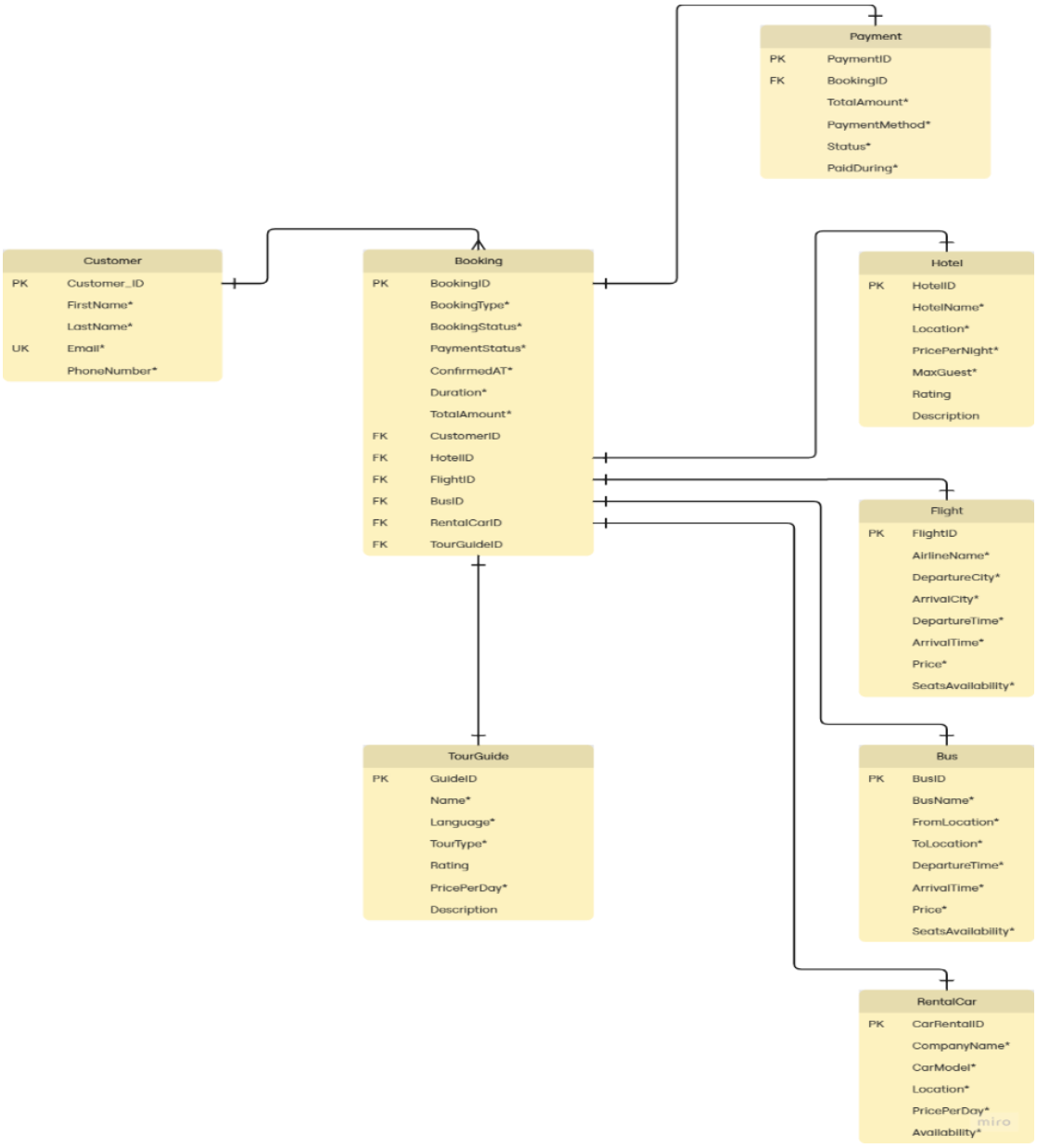
PlanID	Unique identifier for each Plan	Integer	N/A	Primary Key, Auto-increment	N/A	One to Many (Booking)
UserID	Unique identifier for user	Intteger	N/A	Not Null	N/A	Many to One (Booking)
Budget	User's budget preference	Decimal	N/A	Not Null	0.00	N/A
DescriptionText	User input preference trip details	String	200	Not Null	N/A	N/A
Duration	Duration preference of trip	Datetime	N/A	Not Null	N/A	N/A
Destination	Destination set by user	String	100	Not Null	N/A	N/A
CreatedAt	Timestamp of when the plan was created	Datetime	N/A	Not Null	N/A	N/A
UserID	ID of the user who made the booking	Integer	N/A	Not Null	N/A	Many to one (User)
HotelID	FK to Hotel table	Integer	N/A	Foreign Key (Hotel)	NULL	One to One (Hotel)S
FlightID	FK to Flight table	Integer	N/A	Foreign Key (Flight)	NULL	One to One (Flight)
BusID	FK to Bus table	Integer	N/A	Foreign Key (Bus)	NULL	One to One (Bus)

RentalCarID	FK to RentalCar table	Integer	N/A	Foreign Key (RentalCar)	NULL	One to One (RentalCar)
TourGuideID	FK to TourGuide table	Integer	N/A	Foreign Key (TourGuide)	NULL	One to One (TourGuide)
PaymentID	FK to Payment table	Integer	N/A	Foreign Key (Payment)	NULL	One to One (Payment)

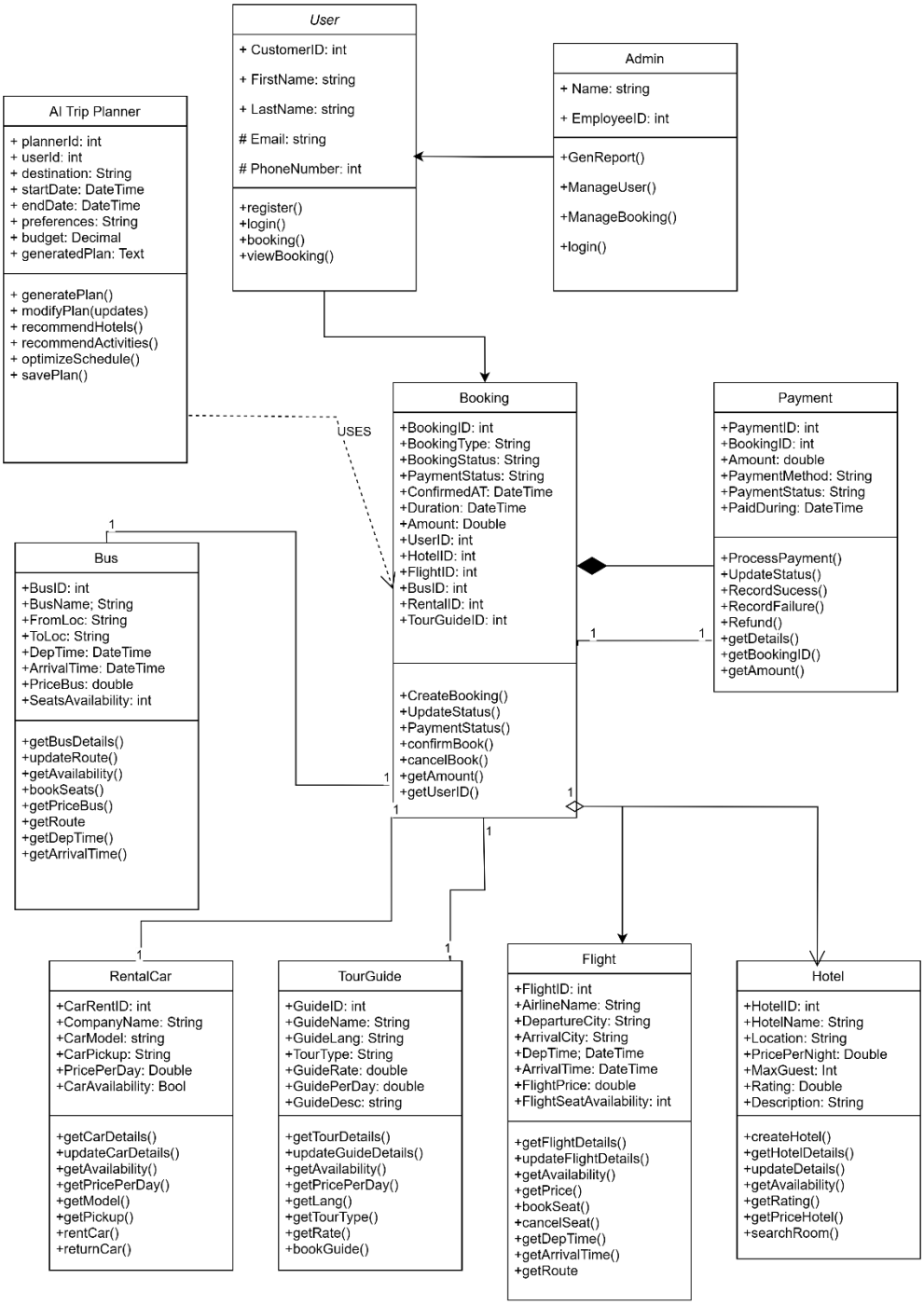
1.8 ERD Diagram



Physical ERD Diagram



2.0 Class Diagram



The user class represents the user interaction with the booking system. It stores personal information like the userID, name, email, and phone number. The user can make multiple types of bookings through the booking class where it is a one-to-many relationship. The user class also enables the users to register, login, and create bookings according to their choice itself.

Then, the admin class controls the whole system where it allows the admin to access rights to manage the system. Although the admin class share a common action with the user class, an admin can manage multiple users and bookings implying it as a one-to-many relationship that allows them to oversee all the system operations.

The booking class is the main class that allows users to book and confirm their reservation. The class stores information like the booking status for each separate services and it is associated with one payment window. The booking class also has a one-to-one relationship with the services available to allow each user to book what they prefer easily. Additionally, each Booking is also associated **one-to-one** with each service class (Hotel, Flight, Bus, TourGuide, RentalCar), ensuring clear tracking of which services a user has selected.

The payment class stores all the booking information and amount of the booking that is prepared by the booking class. It records all the financial transactions details for a booking that includes the paymentID, amount of the booking, paymentMethod, paymentStatus, and the bookingID. A payment is directed with one booking signifying a one-to-one relationship with the booking class.

The AI Trip Planner class automates travel planning with attributes like destination, budget, and preferences. The AI Trip Planner uses these inputs to recommend bookings and generate complete travel plans. It connects directly to the Booking class, allowing users to transform an itinerary into confirmed bookings. It also maintains an association with the User class (one-to-many), where each user can create and manage multiple AI-generated plans.

The services that the system provides are hotel, bus, tour guide, flight, and car rental bookings. Each of the class contains specific information for each service that allows the users to book their own preference. The information of the booking of each service will then be stored back to the booking class to create the final booking information of the user and will display the amount of the booking for the user.

3.0 Use Case Diagram

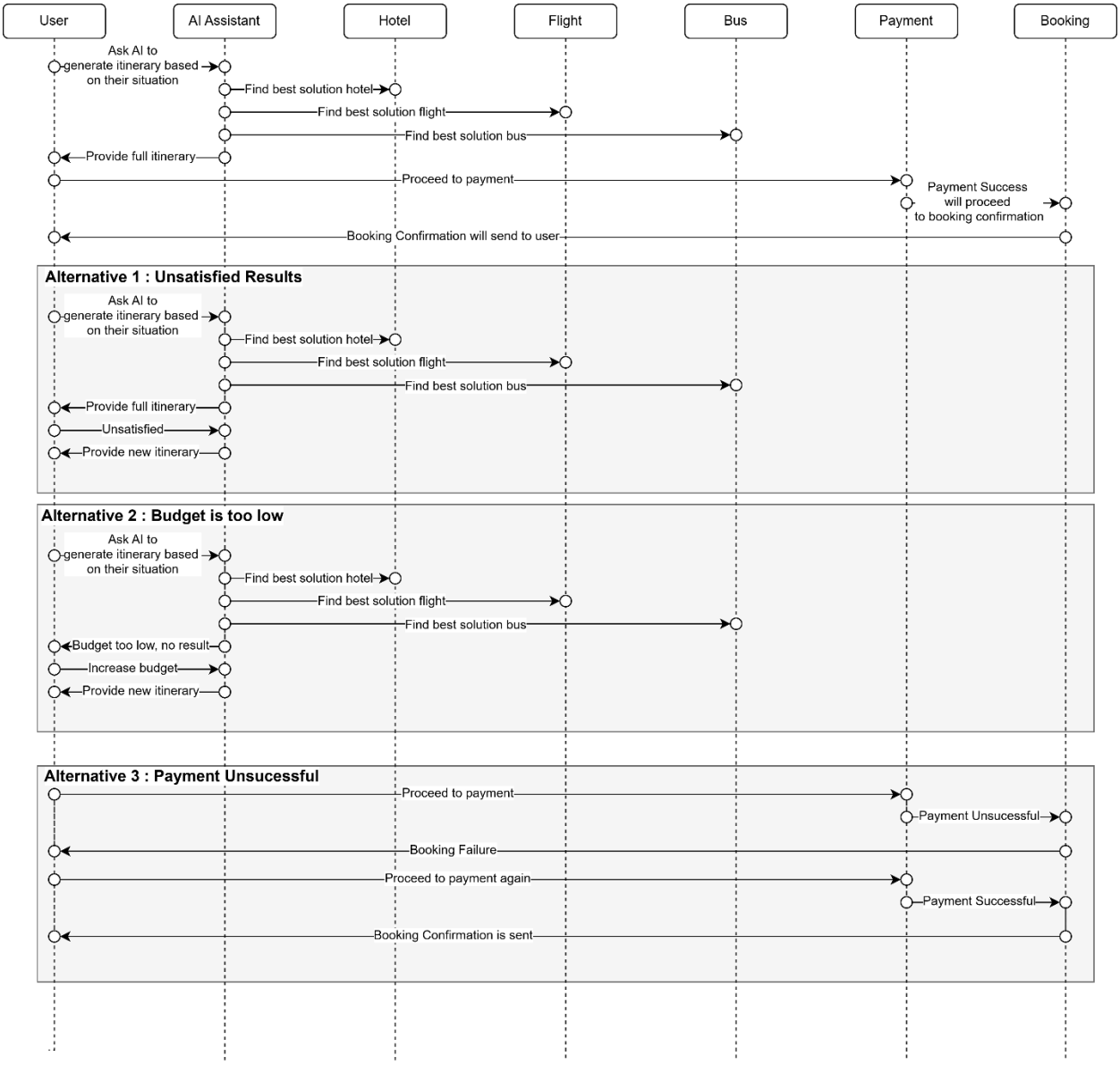


The use case diagram shows there are 4 main actors involved in the booking.com system, which is the user, admin or staff, payment gateway and book system. The users

are the users of Booking.com. They must log in first before accessing the application. After that, they can start browsing for hotels, flights and others on their own. They can also choose to ask AI trip assistants by entering their budget, duration and number of people to generate a great trip plan for them. If they are satisfied with the results from their own search or AI assistant, they can make a booking and make their payment. They can choose to pay with their debit or credit card or choose to pay later. After that, they will receive the booking confirmation once the payment has been processed by the payment gateway. The booking system will record in their database. In addition, they can manage their booking whether to find and adjust the booking or cancel the booking. At the same time, the admin can also manage the booking if there are changes made by the hotel, flight and bus company. Furthermore, they can also use the e-hailing function to call the taxi or use the car rental function to rent their car at their destination. Lastly, they can book a tour guide to take them to travel across the city or country.

4.0 Sequence Diagram

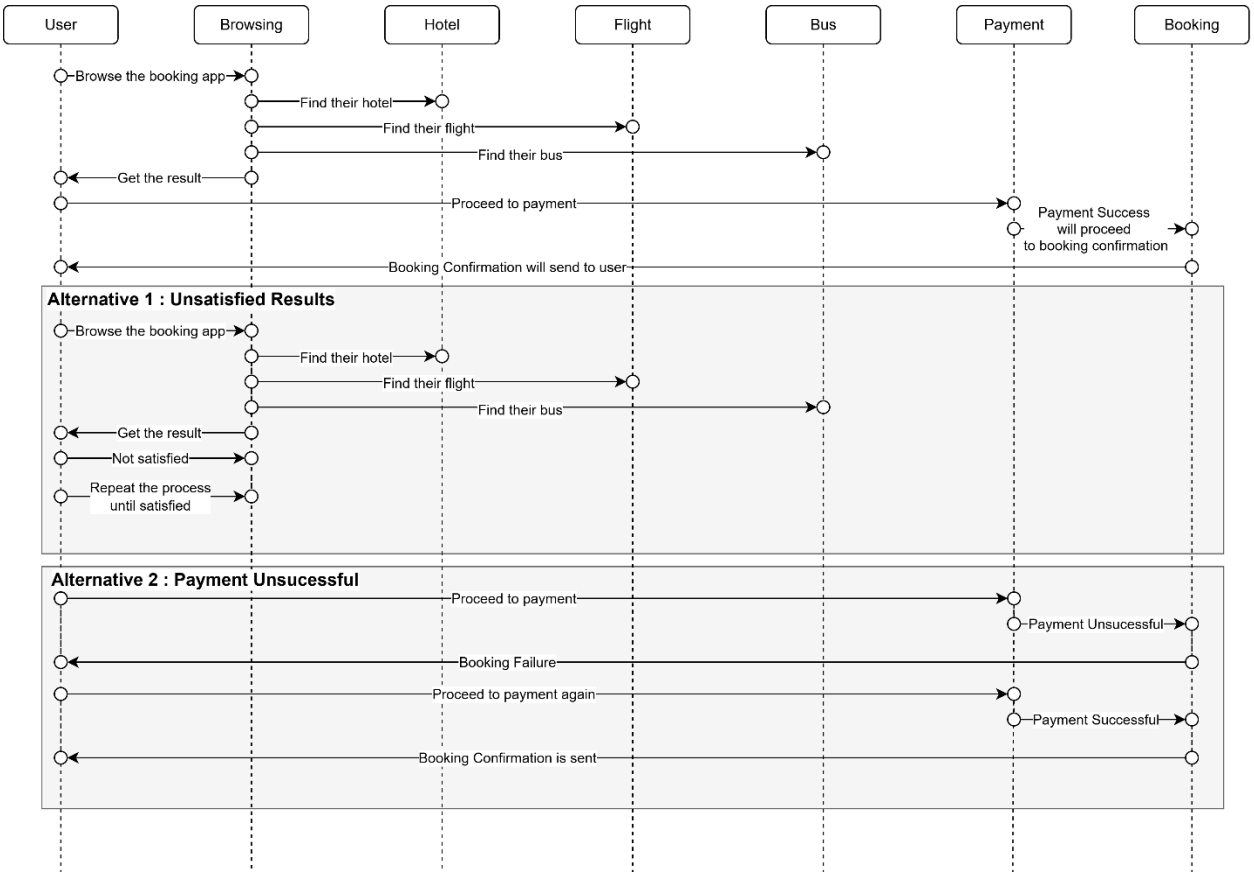
Sequence Diagram 1 – AI Trip Planner



The process begins when the user asks the AI assistant to generate a personalized travel itinerary based on their situation such as preferences, budget, or timing. The AI assistant will help to searches for hotels, flights, and buses to find the best possible combinations and shows a full itinerary. If the user is satisfied, the user will proceed with the payment to book them. After the payment is successful, the system confirms the booking and sends a confirmation to the user.

However, if the user is unsatisfied with the itinerary, the AI assistant will search and provide new itinerary. In cases where the user's budget is too low to find any suitable travel plans, the AI notifies the user to adjust their budget, then AI provides new searches. If the payment process fails, the system informs the user of the booking failure and attempts the payment again until successful, after which the booking is confirmed and sent to the user.

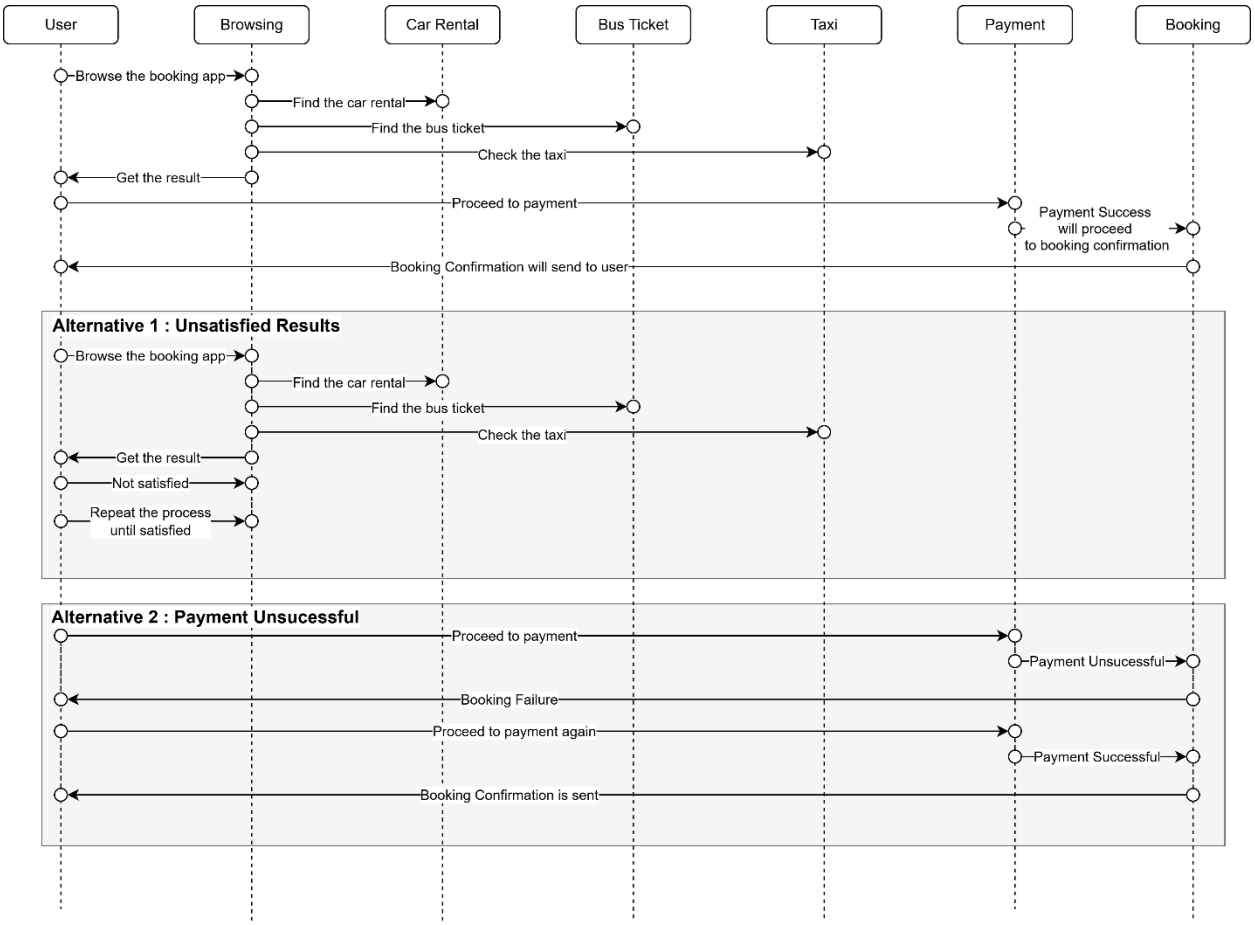
Sequence Diagram 2 – Manual Booking



The user begins by browsing property, searching for flights, bus or other services to book for their vacation. When users are satisfied with the result, they can proceed to their payment via multiple options. When the payment is successful, the system will complete their booking and confirmation page will be sent to the user.

If the users are unsatisfied with their result, they can return to browsing listing to continue their search until they find an alternative that meets their requirements and satisfaction. If the payment fails, the system will inform the user with a notification stated that the booking attempt fails and require users to retry the payment until successful, then the booking is confirmed, and a notification will be sent to the user.

Sequence Diagram 3 – Transportation Booking



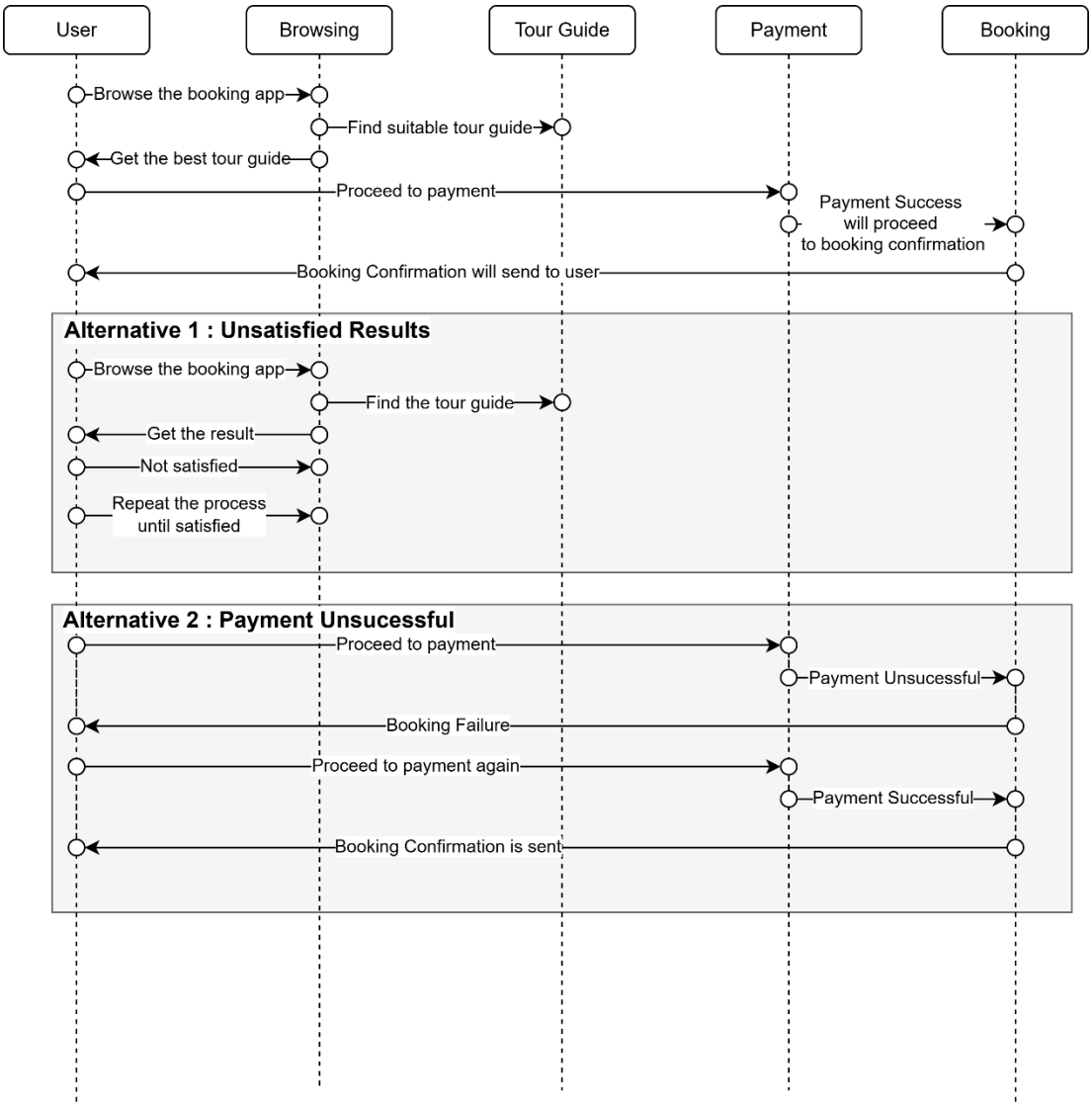
This sequence diagram shows how the interactions of any user on the improved Booking.com platform occurs with the help of different elements throughout a usual trip making and booking procedure. The primary use case starts with the user using the booking application. The browsing component within the system initiates the search of the services, one at a time by querrying the car rental, bus ticket, and taxi modules. After retrieving all the results of the services, the results are sent back to the user to view the same. In case of customer satisfaction with the provided variety, he or she moves to payment. When the payment module returns a successful payment the booking system will create a booking confirmation and send it back to the user thus completing the transaction.

There are also two additional flows shown in the diagram. With the Alternative 1 (Unsatisfied Results), the user explores the app and obtains the results of transportation searches. Nevertheless, in case the user fails to be content with the result, he or she

repeats the process of browsing. This loop is repeated until it comes up with satisfactory results upon which the user would proceed to payment. In Alternative 2 (Payment Unsuccessful), the user clicks on payment when they have already chosen their favored services but the first attempt to pay does not work. When the system detects a failure, the user is informed, and he repeats the payment. The second time it works out, and a booking confirmation is created, which is passed to the user.

In general, this sequence diagram presents an effective and convenient booking process. It includes the normal processes and error processing in the system like results dissatisfaction and unsuccessful payment, to give a seamless and sustainable user experience.

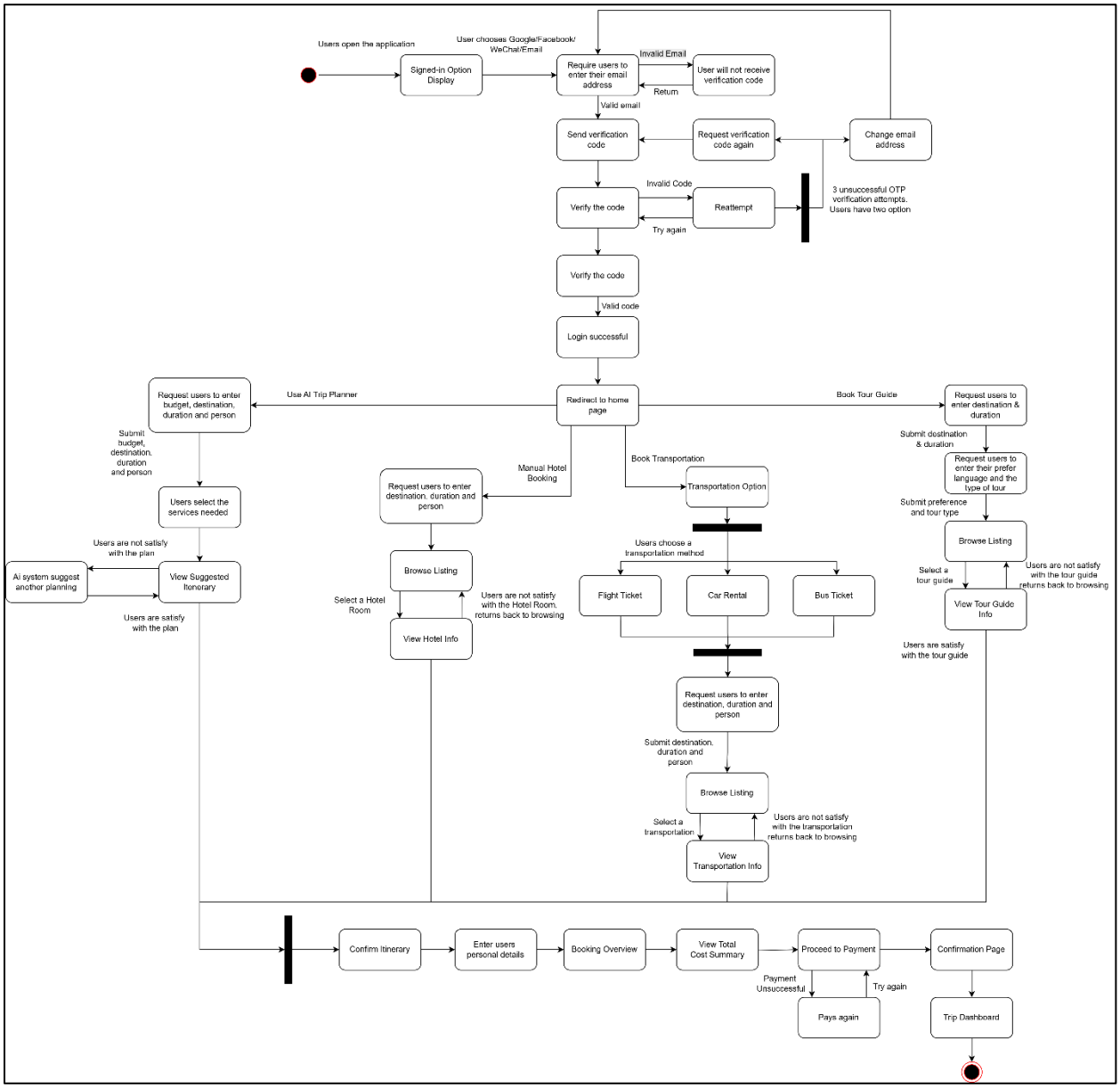
Sequence Diagram 4 – Tour Guide Booking



The user started to browse Booking.com application to look for a suitable tour guide to bring them play around a specific city. If they have founded the best tour guide to choose, they will proceed to book and make their payment. After the payment successful, the booking confirmation will be sent to the user.

If they do not find the satisfied tour guide; they will continue to seek for the best tour guide that fulfil their requirements. If they payment is unsuccessful, the system will ask user to make the payment again until the payment is successful.

5.0 State Transition Diagram



The Initial step of the system requires users to log in into Booking.com application. Our system provides various login options, which include Google, Facebook, WeChat or Email Address. One of the most used methods is email address. In this process, users must enter their email address to receive a verification code. The verification code will be sent strictly to the email address entered. Therefore, users must ensure to enter their address correctly, or else users would need to re-enter their valid address. Once users

receive and enter the code, the system will verify it. If the code is valid, login will be successful. If the code is invalid, users are allowed to retry up to three attempts. After the third failed attempt, the system will reject the current code, and users will be given two options which are request a new verification code or return to the email input step to provide a different email address.

After they log in successful, the users will be redirected to the home page. If they want to make a plan but not sure how to begin, they can use the AI-Driven Trip Planner feature to help them plan a full itinerary and package. The system will request users to enter their budget, destination, duration and number of travellers for the trip. Next, users could select the service they needed for the trip such as a full itinerary (including flight, hotel, transportation booking and tour guides booking), hotel booking, flight booking and/or bus transportation. Once services are selected, the AI system will provide a suggestion based on user's requirement, if users are not satisfied with the suggestion, they can request for a new plan until users are satisfied with the suggestion overview.

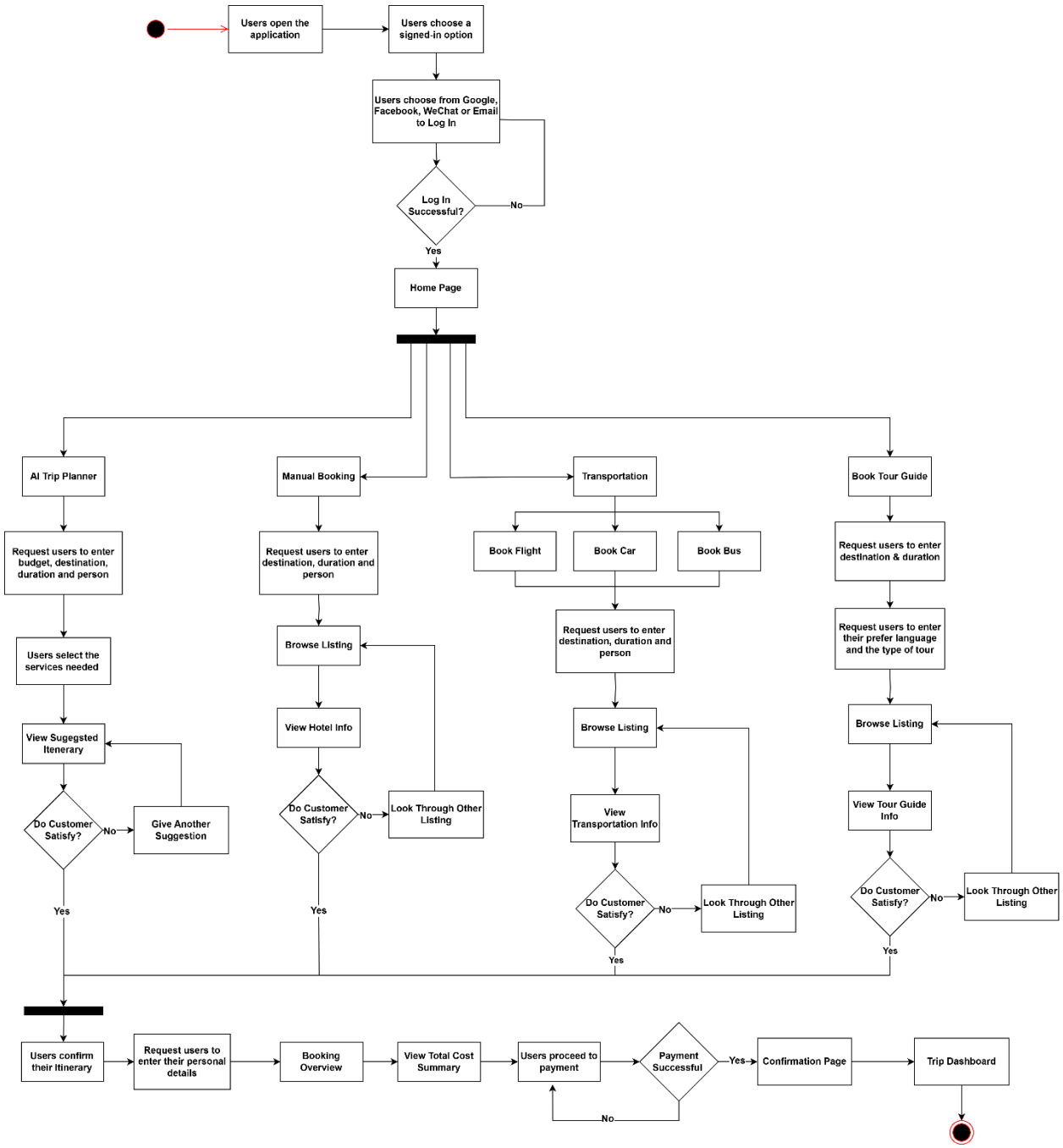
If users want to book a hotel room, they can use the manual hotel booking system. The system will prompt users to enter their destination, duration and number of travellers for the trip. After filling in the request, a list of relevant hotels will be display for users to browse. Users can view each hotel's details for more information. If they are not satisfied with a particular option, they can return back to the list and continue browsing until they find a suitable hotel.

If users want to book transportation, they can choose from three service which are flight, car rental and/or bus ticket. Once a service is selected, the system will prompt users to enter their destination, duration and number of travellers for the trip. After filling in the request, a list of relevant flight/ car rental/ bus option will be display for users to browse. Users can view for detailed information about each option. If they are not satisfied with a particular option, they can return to the list and continue browsing until they find a suitable transportation option.

Users can use the tour guide booking system to find a suitable tour guide for their trip. When the system is selected, the system will prompt users to enter their destination, duration, prefer language and the type of tour for the trip. After filling the request, a list of relevant tour guide will be display for users to browse. Users can view for detailed information about each guide including their rating, review and other relevant details. If users are not satisfied with a particular option, they can return to the list and continue browsing until they find a suitable tour guide.

When satisfied, users can confirm the itinerary. At this stage, the system will prompt users to enter their personal details to ensure the booking is associated with related person. Users can then review all details in the Booking Overview section before proceeding to payment. If the payment fails, the system will prompt users to retry until it is successfully processed. Once the payment is completed, the system will redirect users to a confirmation page, where they can view and manage their trip details via the Trip Dashboard.

6.0 Activity Diagram



The Initial step of the system requires users to log in into Booking.com application. Our system provides various login options, which include Google, Facebook, WeChat or Email Address. If the log in was not successful, the system will request users to try again.

Upon successful login, users are directed to the home page. If users are planning for a vacation but don't know where to start, they can try out the AI-Driven Trip Planner feature for help. To begin with, the system will ask users to provide details of their trip such as their budget, destination, duration and number of travellers. Next, users can choose different services they would require during the trip such as a complete itinerary (including flight, hotel, transportation, and tour guide bookings), individual hotel reservation, individual flight booking, and/or individual bus transportation. Once services are selected, the AI system will generate a suggestion based on the user's needs. If users find this suggestion unsatisfactory, they can request a new plan until they are content with the suggestion overview.

For booking a hotel room, users can utilize the manual hotel booking system. Users will be prompted by the system to provide their destination, trip duration, and number of travellers. Once users have completed the request, a list of pertinent hotels will be shown for them to browse. Users can access each hotel's details for further information. If they don't find the current hotel to be satisfied, they can go back to the list and keep browsing until they locate an appropriate hotel.

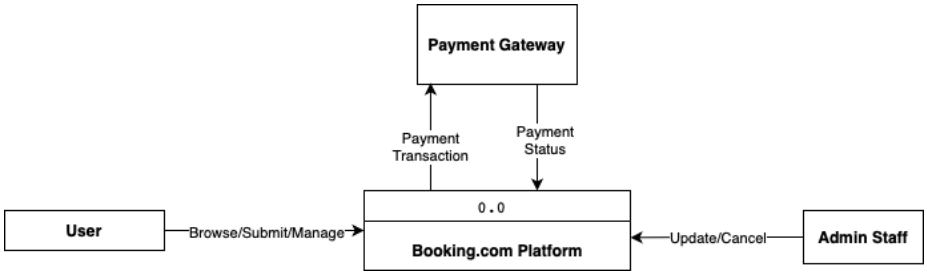
If users want to book transportation, they can choose from three service which are flight, car rental and/or bus ticket. The system will prompt users to enter their destination, duration and number of travellers for the trip. After filling in the request, a list of relevant flight/ car rental/ bus option will be displayed for users to browse. Users can view for detailed information about each option. If they are not satisfied with the current option, they can return to the list and continue browsing until they find a suitable transportation option.

The tour guide booking system allows users to find a suitable tour guide for their trip. When the system is selected, the system will prompt users to enter their destination,

duration, prefer language and the type of tour for the trip. After filling the requirements, a list of relevant tour guides will be shown to the users to browse. Users can access the tour guide detail information such as their rating, reviews, and other pertinent details. If users are dissatisfied with a specific option, they can go back to the list and keep browsing until they locate a suitable tour guide.

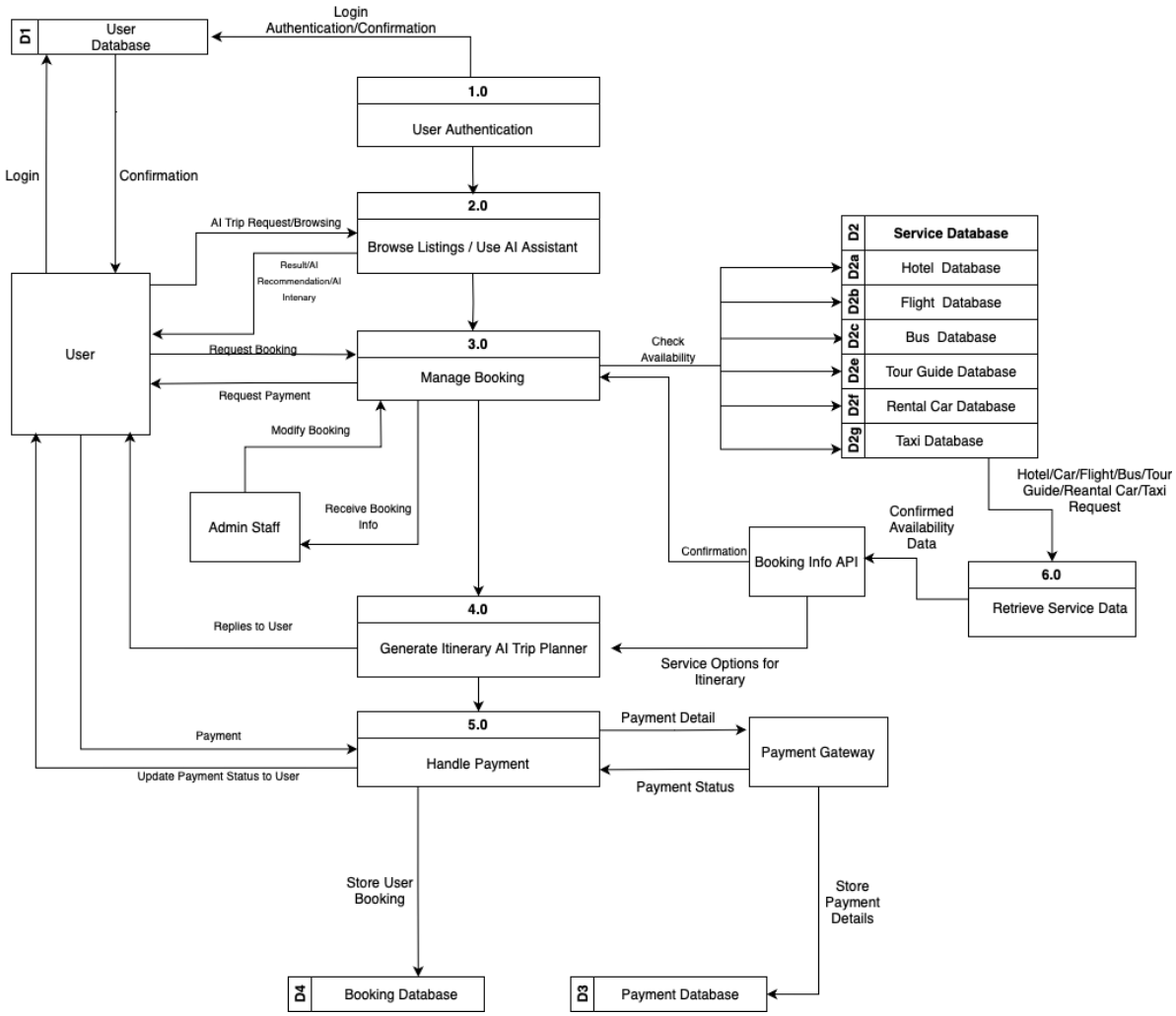
When satisfied, users can confirm the itinerary. At this stage, the system will prompt users to enter their personal details to ensure that the booking is linked to the corresponding individual. Before proceeding with payment, users can review their details in the Booking Overview section again to ensure no mistakes are made. If the payment fails, users will receive a notification stated that the booking attempt fails and users need to retry the payment until it is successfully processed. Upon payment completion, users will be redirected to a confirmation page by the system, where they can view and manage their trip details through the Trip Dashboard.

7.0 DFD Diagram



This is a context diagram of the communications between the Booking.com Platform (Process 0.0) and three external entities, the User, the Payment Gateway and the Admin Staff. The users use the platform to make browsing, submissions, and manage their bookings and the platform interacts with the Payment Gateway through relaying payment transactions and retrieving the payment status confirmations. Admin Staff use the system to correct records of booking or cancel listing. The diagram is at a conceptual level of the main external data exchanges of the system and forms the basis of more detailed process modelling.

Level 1 DFD – Decomposition of Booking.com Platform (Process 0)



The Level 1 Data Flow Diagram (DFD) of the Booking.com Platform gives an elaborate breakdown of the major process of the system (Process 0) into 6 sub processes connected with each other. It shows the interaction of the system with other external entities, data stores and other components of the system to provide an improved experience of travel planning and booking. The flowchart indicates the collaboration of the AI assisted planning of itineraries and unified travel experiences in the form of booking hotel rooms, traveling commutes and tour arrangements.

Process 1.0 starts with user authentication. In case a user tries to log on, the user database (D1) is consulted to determine whether the credentials entered by the user are correct. On successful authentication, the user is then notified on successful authentication and proceeds to access the core features of the application. After the success of the authentication process, the users will be able to search the listings or order AI-made itineraries through Process 2.0, ""Browse Listings / Use AI Assistant," Through this process, the app grasps the preferences of the user and either displays them manually, or calls the AI engine, to give personalized recommendations on travel. The findings in form of accommodation suggestions to complete schedules are transmitted back to the user interface.

Once a user has decided that they are ready, they request a booking using Process 3.0, "Manage Booking." This one deals with receipt of booking information and verifies available services of hotels, flights, buses, tours and car rents among others. Availability requests are communicated to Process 6.0, Retrieve Service Data, which communicates with a set of so-called service databases (D2a-D2f), including Hotel Database, Tour Guide Database, and Rental Car Database, instead of querying each of these databases individually. The availability data is confirmed and sent to the Booking Info API which is then used in the itinerary or during the booking confirmation.

The process 4.0, Generate Itinerary AI Trip Planner, takes the confirmed availability information and creates an intelligent travel plan based on the preferences of the user. The remaining plan is posted back to the customer and the entire travel planning service is automated and customized. In case the user decides to accept the itinerary or create any reservation, process 5.0, the process of handling payment is executed. This step transfers details of payment through to the external Payment Gateway and returns the transaction pending status which is then updated in the system. Achieved payment causes data to be inserted into Booking Database (D3) and Payment Database (D4) and notifies the user about the result.

The system also facilitates communication with Admin Staff. Admin Staff can get the information that shows booking and do the necessary change, or corrections unlike in the past with Process 3.0 which gives room to operational flexibility. All the flows of the data are labelled correctly and are routed correctly through system processes and are not directly accessible to users and external systems- compliant to the DFD rules and best practices.

In general, this Level 1 DFD represents a modular view, logically structured version of the improved Booking.com. It demonstrates explicitly how so-called user interactions, back-end service integration, and AI based features could combine to provide a smooth travel planning process. The diagram helps in traceability, scaling of the system and easier comprehension of the connection between the components that make the system achieve its user and administrative functionalities.